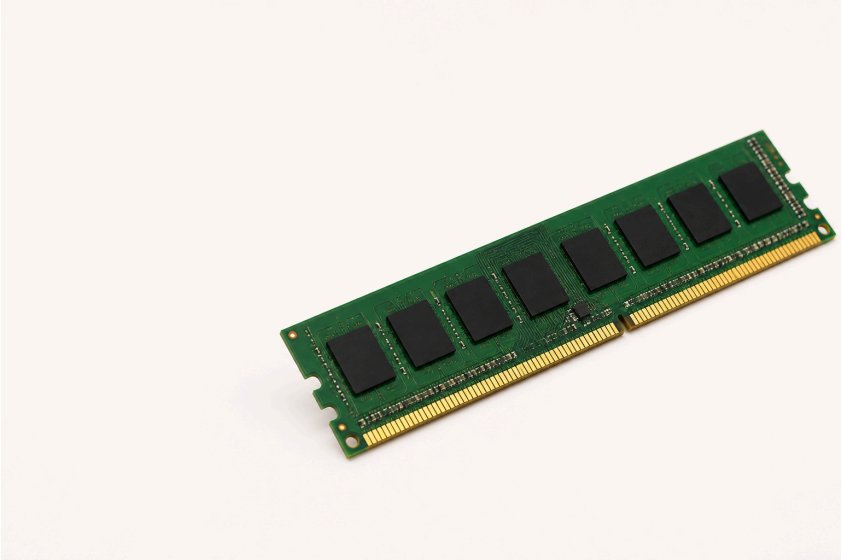
Lecture 2: Data Representation I — Bits, Bytes & Integers

Session 2 · Mon Aug 31, 2026

From Physical Memory to One Byte

That is a **RAM module** — physical hardware, many chips, not one byte.

Memory is **byte-addressable**: every address names one **8-bit byte**, and a byte is the smallest unit a C program can address on its own.

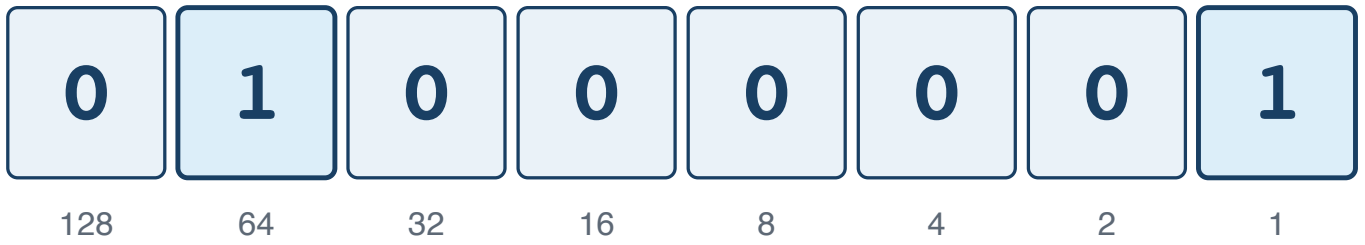


One byte = 8 bits. Today is about what fits in one, and how to work on it.

One Byte, Two Readings

Start with one byte: eight switches that are each either 0 or 1.

One byte: the bit pattern for 65



64 + 1 = 65

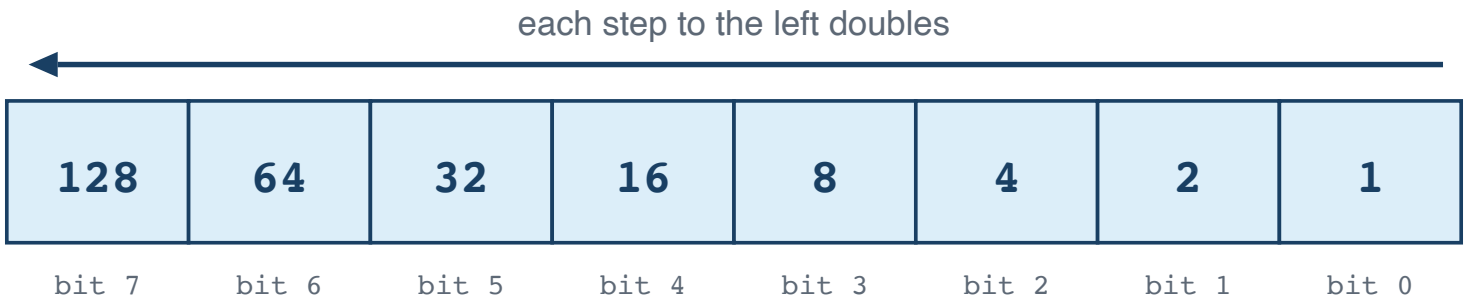
as text (ASCII): A

The computer stores only the bits; the program decides how to read them.

Counting in Binary

Why does `01000001` mean 65? Because each place is worth twice the one to its right — the same idea as decimal's ones, tens and hundreds, with 2 instead of 10.

Each place is worth twice the place to its right



Turn a bit on and you add its place value.

128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255

Counting Up

Counting works exactly the way it does in decimal — when a column runs out of digits, it rolls over and carries into the next one:

```
0 = 0000    3 = 0011    6 = 0110    9  = 1001
1 = 0001    4 = 0100    7 = 0111    10 = 1010
2 = 0010    5 = 0101    8 = 1000    11 = 1011
```

Each extra bit doubles the patterns available: 8 bits give **2^8 = 256**. So an unsigned byte runs 0–255, and an N-bit unsigned value runs 0 to `2^N - 1` — its **UMax**, modulo `2^N`.

Writing the Same Number Three Ways

A number does not change when you write it differently. C lets you say which notation you are using with a **prefix**:

Written	Prefix means	Value
`42`	no prefix — plain decimal	forty-two
`0b101010`	`0b` — read the digits as **binary**	forty-two
`0x2A`	`0x` — read the digits as **hexadecimal**	forty-two

All three are the same value to the compiler. You pick whichever makes the thing you are doing readable: decimal for counting, binary for seeing bits, hex for writing bytes compactly.

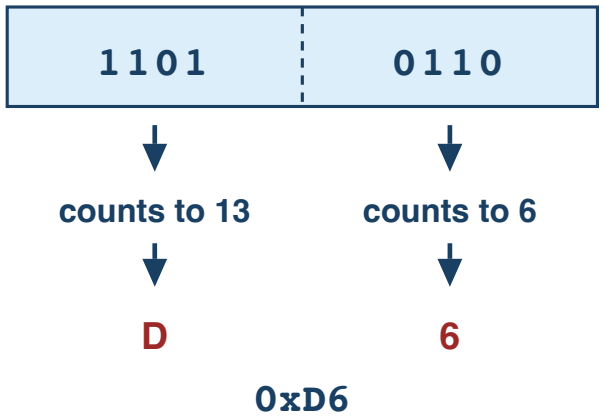
> **Note**
>
> `0b` is not standard until C23. GCC and Clang accept it, and these slides use it to show bit patterns, but in ISO C17 write `0x2A` or `42`.

Why Hex Fits Bytes So Well

Eight ones and zeros is easy to miscount. Four bits count 0–15 — exactly one hex digit — so one byte is always two digits, and you can still read the bits straight off the page.

One byte = two hex digits

split the 8 bits into two halves of 4



4 bits count 0 to 15,
so we need 6 extra digits:

```
10 = A
11 = B
12 = C
13 = D
14 = E
15 = F
```

0 through 9 stay as they are.

Characters Are Just Numbers

The rule that turns 65 into `A` is **ASCII** — the American Standard Code for Information Interchange — a table agreed in the 1960s that gives every character a number from 0 to 127.

Characters	Codes	Worth knowing
`'0'`–`'9'`	48–57 (`0x30`–`0x39`)	consecutive, so `c - '0'` gives the digit
`'A'`–`'Z'`	65–90 (`0x41`–`0x5A`)	consecutive
`'a'`–`'z'`	97–122 (`0x61`–`0x7A`)	exactly 32 more than the capitals

```
char c = '7';
int  d = c - '0';                          // 7
int  is_digit = (c >= '0' && c <= '9');  // 1
```

In C the literal `'A'` **is** the number 65. Nothing clever is happening.

The Same Ideas in Python

C

```c
int x = 65;
printf("%d\n", x);      // 65
printf("%c\n", x);      // A

unsigned a = 12, b = 10;
printf("%u\n", a & b);  // 8
printf("%u\n", a << 2); // 48
```

Python

```python
x = 65
print(x)            # 65
print(chr(x))       # A

a, b = 12, 10
print(a & b)        # 8
print(a << 2)       # 48
```

The operators are the same. The difference is the box: a C value has a fixed width, and a Python integer just grows.

Bit-Level Operations

For these bit patterns, C provides six bitwise operators:

Op	Name	Example
`&`	AND	`0b1100 & 0b1010 = 0b1000`
`|`	OR	`0b1100 | 0b1010 = 0b1110`
`^`	XOR	`0b1100 ^ 0b1010 = 0b0110`
`~`	NOT	`~0b1100 = 0b1111...11110011` (every bit flips, including the leading zeros)
`<<`	Left shift	`0b0011 << 2 = 0b1100`
`>>`	Right shift	`0b1100 >> 2 = 0b0011`

> ⚠ **Warning**
>
> Right shift of signed integers is **implementation-defined** in C. GCC uses **arithmetic** right shift (preserves sign bit). Don't rely on this unless you're sure of your compiler.

Bitwise Operations, One Column at a Time

The same two inputs, three separate rules

Every output bit is decided on its own, from just the two bits directly above it.

INPUTS

a	1	1	0	0	= 1100
b	1	0	1	0	= 1010

a & b	1	0	0	0	AND = 1000

1 only where both are 1

a | b	1	1	1	0	OR = 1110

1 where either is 1

a ^ b	0	1	1	0	XOR = 0110

1 where they differ

De Morgan's Laws

Two rules let you rewrite any mix of &, | and ~ — and they are how Lab 1 asks you to build ^ out of nothing but ~ and &.

Flip the whole thing, and AND becomes OR

```
~(x & y) == ~x | ~y
```

```
~(x | y) == ~x & ~y
```

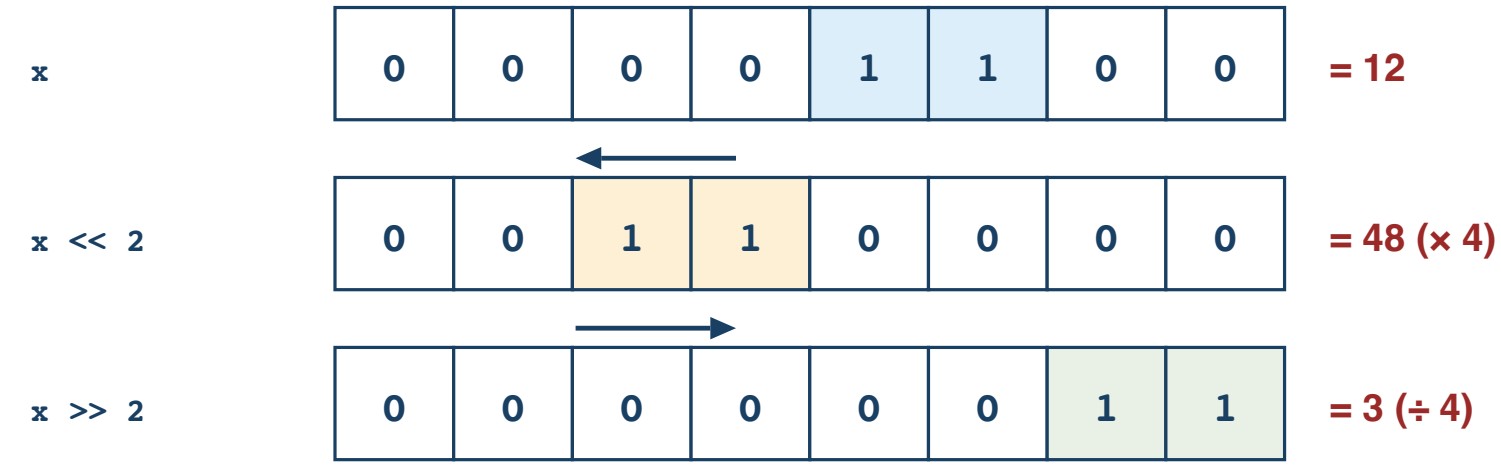
x	y	x & y	~(x & y)	~x | ~y
0	0	0	1	1
0	1	0	1	1
1	0	0	1	1
1	1	1	0	0

same in all four rows

Two inputs give four possible rows, so checking all four is a complete proof.

Shifting Bits

Shifting slides every bit sideways

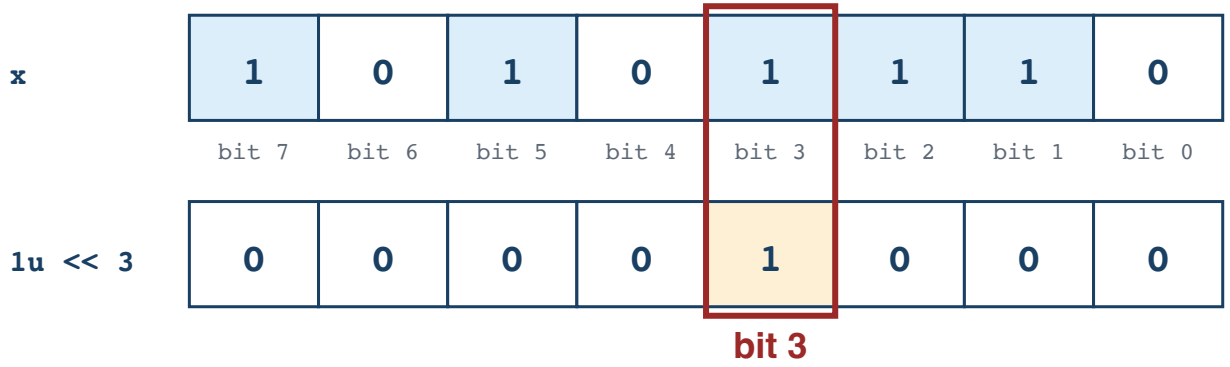


Bits pushed off the end are gone for good; zeros come in behind them.

Shifting left by n multiplies by 2^n. Shifting an unsigned value right by n divides by 2^n.

Test, Set, Clear, Toggle

1u << 3 builds a mask with exactly one bit set



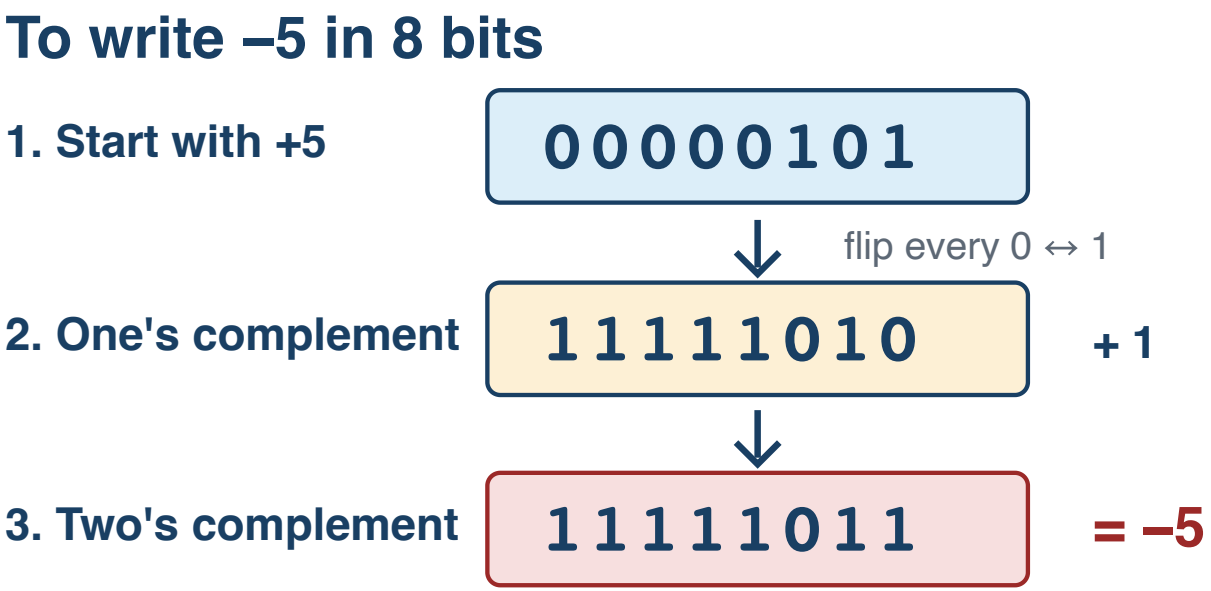
Bit 0 is the rightmost — the ones place. The mask is 0 everywhere except the bit you want.

```
unsigned x = 0xAE;              // 10101110
x |=  (1u << 3);                // SET     — bit 3 becomes 1
x &= ~(1u << 3);                // CLEAR   — bit 3 becomes 0
x ^=  (1u << 3);                // TOGGLE  — bit 3 flips
if (x & (1u << 3)) { }          // TEST    — nonzero when bit 3 is 1
```

Those four are most of Lab 1. Keep the value **unsigned** — shifting into a sign bit is undefined.

What Two's Complement Means

Complement means the part that completes a whole. Flip every bit of x to get its *one's complement*; add 1 to that and you have its **two's complement**, which is how C writes –x:



More 8-bit examples

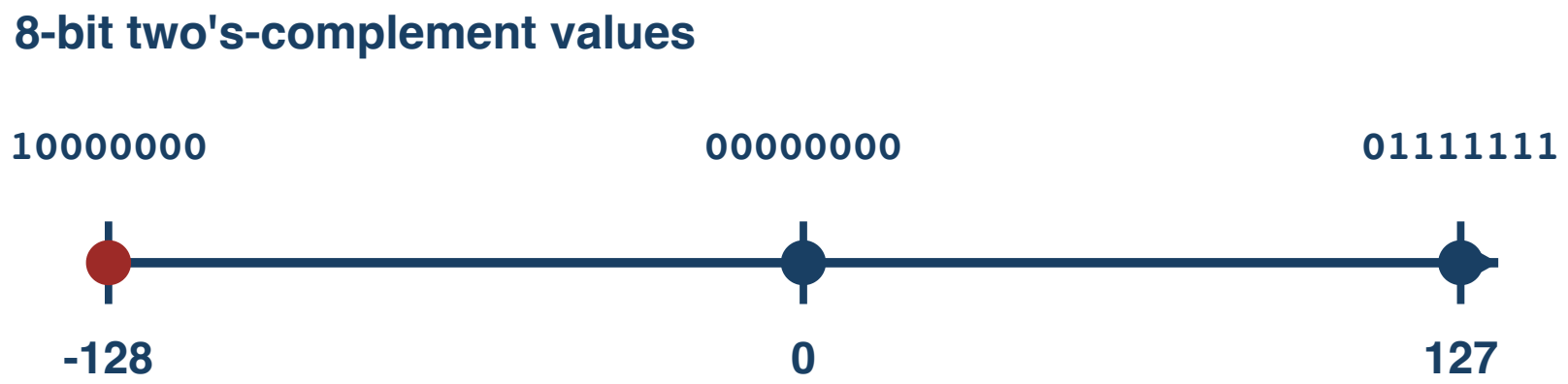
```
+1: 00000001
-1: 11111111

0: 00000000
two's: 00000000
```

Zero stays zero.

A ninth bit is simply discarded — 8 bits keep only their rightmost 8.

The Range It Buys You

An N-bit two's complement covers `–2^(N–1)` to `2^(N–1) – 1`:



There is one more negative value than positive value.

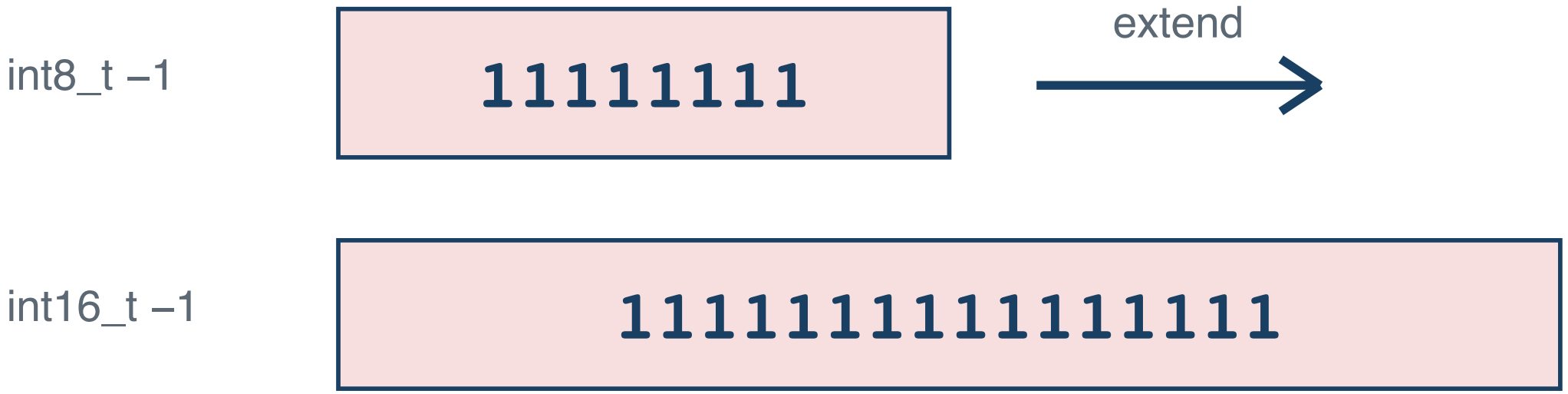
`TMin = –TMax – 1` — there is no positive 128 in 8 bits, so negating `TMin` overflows and is undefined.

Sign Extension

When converting a smaller signed type to a larger one, copy the sign bit:

```
int8_t  x = -1;   // 0b11111111
int16_t y = x;    // 0b1111111111111111 = -1 ✓
```

Copy the sign bit into the new high bits

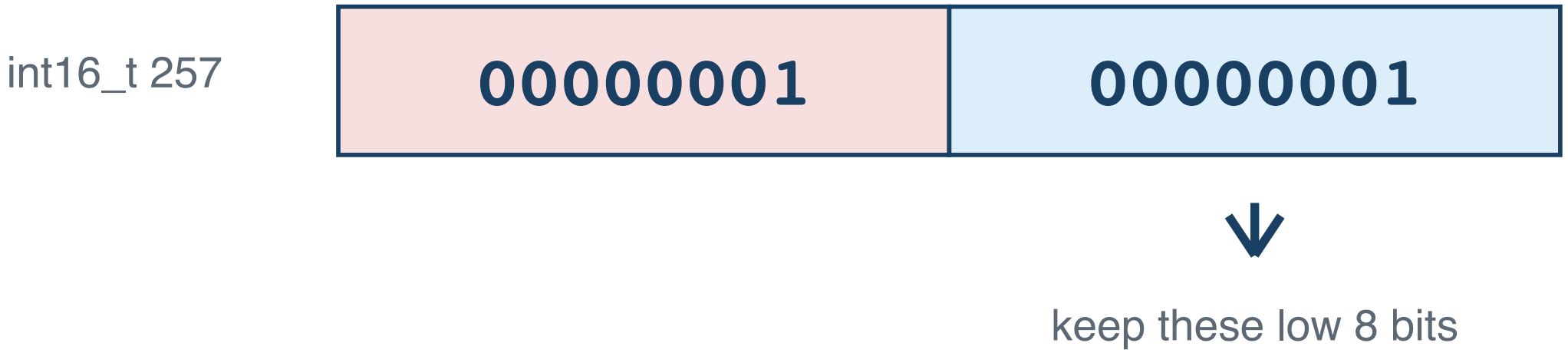


Truncation

When converting larger to smaller, drop the high bits:

```
int16_t  x = 257;
uint8_t  y = x;    // 1: conversion to unsigned is modulo 256
```

Keep the low bits; discard the high bits



int8_t result = 00000001 = 1

Size of Integer Types

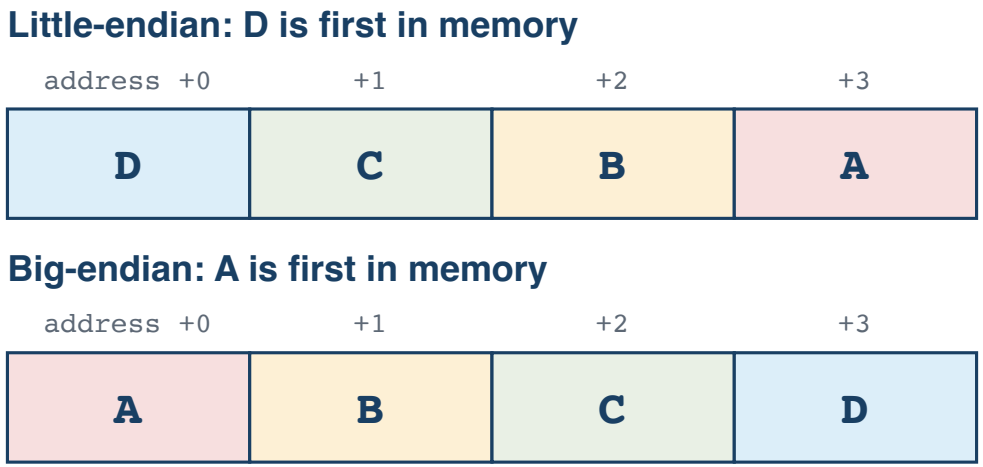
```
printf("char:      %zu\n", sizeof(char));      // 1 C byte (not always 8 bits)
printf("short:     %zu\n", sizeof(short));     // 2
printf("int:       %zu\n", sizeof(int));       // 4
printf("long:      %zu\n", sizeof(long));      // 8 (LP64)
printf("long long: %zu\n", sizeof(long long)); // 8
printf("pointer:   %zu\n", sizeof(void*));     // 8
```

Tip

Use `sizeof` and fixed-width types (`int32_t`, `uint64_t`) when exact sizes matter. Never assume `int` is 4 bytes; use `CHAR_BIT` when bit width matters. Plain `char` is its own trap: whether it is signed is implementation-defined, so say `signed char` or `unsigned char` when it matters.

Byte Ordering (Endianness)

An `int` occupies 4 consecutive bytes, so those 4 bytes have to be laid out in some order. **Endianness** is that order. One question matters: **which byte goes at the lowest address?**




```c
int x = 1;   // ask the machine: which byte of x sits at the lowest address?
if (*(unsigned char *)&x == 1) printf("Little-endian\n");
else                           printf("Big-endian\n");
```

AI Systems Connection

Quantization is one important reason large models can fit on more hardware. A common symmetric int8 scheme represents a block of weights with 8-bit signed values and a scale factor:

```
real_value  ≈  scale × int8_value          scale = max_abs / 127
```

Everything today shows up here. Two int8 values need 16 bits for their product, so a long dot product accumulates into something wider — often `int32_t`. And if a signed weight is loaded as *unsigned* into a wider register, sign extension does not happen, negatives become huge positives, and the answer is quietly wrong.

Try it: quantize a float array with `scale = max_abs / 127`, dequantize, and measure the error.

Practice

1. Write `0xD6` in binary, and `10110010` in hex, without using a computer.
2. What does `x & 1` tell you about `x`? And what does `x & (x – 1)` do? Try both on `0b0110` by hand.
3. Write one expression that clears bit 5 of `unsigned x`, and one that tests whether bit 0 and bit 7 are both set.
4. What is `TMax` for a 32-bit `int`? What is `TMin`?
5. Write a C expression that checks if your machine is little-endian.
6. Is left-shifting a 32-bit `int` by 32 positions defined in C? Why not?
7. Why is `–TMin` not representable in the same signed type?

Reading

- CS:APP §2.1 (Information Storage)
- CS:APP §2.2 (Integer Representations)